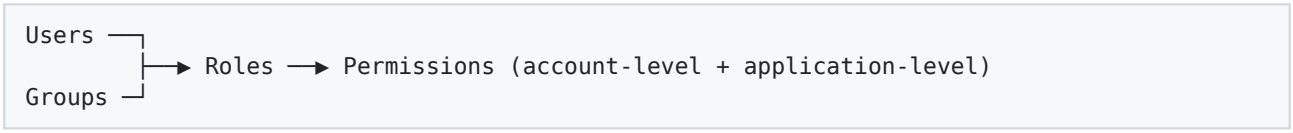
Day 8 — Post-Deployment Verification, RBAC and Revision

Cisco AppDynamics Professional Implementer (500-430 CAPI) · 10-day course · Blueprint 4.1, 4.2, 4.3 (Domain 4 = 5%) + revision of Domains 1–2

Goal: control access with roles and permissions, prove a deployment is healthy, use the Administration Console, and revise the most-tested platform facts.
Verification: every fact checked against the current Splunk AppDynamics on-prem docs (November 2026).

Block 1 — Role-based access control (4.1)



Roles can be given to users or groups. Groups are easier to manage and are how LDAP/SAML mapping works.

Predefined role	What it can do
Account Owner	Manages **users, groups, roles and the authentication provider** + most account permissions — the **only** role that can do user/role administration
Administrator	Day-to-day objects (applications, dashboards, war rooms, scheduled reports). **Can't** manage users, groups or roles
Analytics Administrator	Analytics access: API keys, extracted fields, saved searches, which roles see which data
Applications and Dashboards Viewer	Views all applications and dashboards, can't edit (**former "Read-Only User"**)
Dashboards Viewer	Views custom dashboards only
DB Monitoring User	Views Database Monitoring; can't change collectors
DB Monitoring Administrator	Views Database Monitoring; adds, edits, deletes collectors
Server Monitoring User	Views Server Monitoring; can't configure it
Server Monitoring Administrator	Views and configures Server Monitoring, incl. Service Availability Monitoring

- **Predefined roles can't be edited** — create a **custom role** instead.
- **Application permissions** apply at **Default → Application → Tier**; the lower level wins. Defaults include **Can Create Applications** and View / Edit / Delete.
- Specific permissions: Configure Health Rules, Configure Business Transactions, Configure Agent Properties, **Agent Advanced Operations** (reset agent, thread dumps, debug logs), Configure Monitoring Level.
- **Several roles:** if one role grants a permission and another doesn't, the **grant wins**.
- **LDAP:** any LDAPv3 directory (verified: Active Directory, OpenLDAP 2.4+); **map LDAP groups to roles**; if LDAP is down or a user isn't found, the Controller **falls back to local login** — keep a local admin account.
- **SAML** single sign-on is the other external option.
- The Administration Console **root** user is separate from account users.

Block 2 — Platform health checks (4.2)

Component	Check	Healthy result
Controller	http://<ctrl>:8090/controller/rest/serverstatus	<available>true</available> (200; HA standby 503)
	bin/platform-admin.sh check-controller-health	No issues
	Log in as the initial admin	UI loads, applications visible
Enterprise Console	http://<ec>:9191 → Platform page	All services running
Events Service	bin/platform-admin.sh show-events-service-health	No issues
	https://<es>:9080/_ping	HTTP 200
	EC → Events Service page	All hosts green
EUM Server	http://<eum>:7001/eumaggregator/ping (or /eumcollector/ping)	ping
App agents	Tiers & Nodes → App Agents	Nodes reporting recent metrics
Machine / DB agents	Servers list; Databases → collector status	Reporting, no collector errors
Controller itself	system account (?enableAccounts=true); disk latency alerts	Normal memory trend; write latency ≤ 3 ms
Licenses	License usage page	Enough units for the agents deployed

Component	Log
Controller	<controller_home>/logs/server.log · db/logs/database.log
Enterprise Console	platform-admin-server.log
Events Service	Collect with retrieve-events-service-logs
Java Agent	<agent_home>/ver<version>/logs/<node>/agent.log
.NET Agent	%ProgramData%\AppDynamics\DotNetAgent\Logs

Block 3 — The Administration Console (4.3)

- URL: http(s)://<ctrl>:<port>/controller/admin.jsp — log in as **root**. Another user logged in → **Access Denied**: log out or use a private window.
- **Controller Settings:** eum.cloud.host · eum.beacon.host · eum.beacon.https.host · eum.mobile.screenshot.host · appdynamics.on.premise.event.service.url/.key · metrics.min.retention.period (4 h) · metrics.ten.min.retention.period (48 h) · metrics.retention.period (365 days).
- **Account Settings:** accounts, licenses, access keys; multi-tenant → each account's EUM License Key and EUM Account Name.

Task	How
Change root user password	Administration Console → user profile → My Settings → My Preferences → Change Password
Reset lost root password	bin/controller.sh login-db → update user set encrypted_password = sha1('<new>') where name='root'; → restart app server (hash upgrades to PBKDF2 at next login)

Task	How
Change MySQL root password	**EC → Configurations → Controller Settings → Database Configurations** (immediate), or manually start-db insecure → ALTER USER

Block 4 — Revision: most-tested platform facts (Days 2–5)

#	Fact	Day
1	Primary sizing driver = metric ingestion rate; second = number of agents	2
2	Demo 10K/5 · Small 50K/50 · Medium 1M/2,000 · Large 5M/15,000 (Linux only)	2
3	Disk write latency ≤ 3 ms; no NFS	2
4	Tenancy: single → multi OK; multi → single never	2
5	Events Service: 1 node or ≥ 3; heap ½ RAM, 7–31 GB; SSD	2
6	EUM Demo = same host as Controller; Production = own host + own MySQL	2, 5
7	EC installer platform-setup-64bit-linux.sh -q -varfile; EC 9191, EC DB 3377, Controller DB 3388	3
8	Start database first, then app server; stop in reverse	3
9	Retention defaults 4 h / 48 h / 365 days	3
10	SSL: keystore.jks, alias s1as, changeit, 8181; chain error → cacerts.jks	4
11	HA: active-passive, identical Linux, activate-ha-modules; > 3 days incremental replication; > 7 days revive	4
12	Disable auto-failover before restarting the primary (Watchdog fails over after 5 min)	3, 4
13	Upgrade order EC → Events Service → EUM → Controller; EC doesn't back up MySQL	4
14	Controller→ES key ad.accountmanager.key.controller; EUM→ES key appdynamics.es.eum.key	5
15	ES upgrade failed → useCheckpoint=true; EUM upgrade → keep EUM MySQL running	5

Lab (~45 min)

1. **Custom role (SaaS):** "PetClinic Viewer" with View on PetClinic only, assigned to group "QA"; add a test user; confirm only PetClinic is visible and read-only.
2. **Combined roles:** give the test user a second role with Configure Health Rules on PetClinic. Can they now edit health rules? Why?
3. **Health rule + policy:** average response time > 2 s on PetClinic, with a policy that sends an email.
4. **Paper:** a 10-step post-go-live verification checklist for the bank platform (HA Controller, 3-node Events Service, EUM Server, agents).

Day 8 Quiz — with answer key

Cover the answer lines and test yourself first.

Q1. Only predefined role that can add/edit users, groups, roles and the authentication provider?

A. Administrator
B. Account Owner
C. Analytics Administrator
D. Server Monitoring Administrator

Answer: B — Administrator handles day-to-day objects, not user administration.

Q2. Team needs Administrator rights minus deleting applications — what do you do?

A. Edit the Administrator role
B. Create a custom role
C. Give Account Owner
D. Change admin.jsp
Answer: B — Predefined roles can't be edited.

Q3. App-level View on PetClinic, tier-level Edit on Payments — which applies to Payments?

A. Application setting
B. Tier setting (lower overrides higher)
C. Default
D. Neither
Answer: B — Default → application → tier; lower wins.

Q4. LDAP server is down — what happens to logins?

A. Nobody logs in
B. Fallback to local authentication; keep a local admin
C. SAML takes over
D. Only root via UI
Answer: B — The docs recommend keeping a local administrator account.

Q5. Which HTTP check confirms the Events Service is healthy?

A. https://<es>:9080/_ping → 200
B. /eumaggregator/ping
C. /controller/rest/serverstatus
D. :9191
Answer: A — Also show-events-service-health or the EC page.

Q6. Change the Controller MySQL root password in the EC?

A. admin.jsp → Account Settings
B. EC → Configurations → Controller Settings → Database Configurations
C. EC → Events Service page
D. Reinstall only
Answer: B — Takes effect immediately.

Q7. Two measurements to estimate Controller requirements?

A. Users + dashboards
B. Metrics/min + agents reporting
C. Health rules + tiers
D. Disk + applications
Answer: B — Prep-guide sample answer.

Q8. Platform upgrade order?

A. Controller → ES → EUM → EC
B. EC → Events Service → EUM Server → Controller
C. EUM → Controller → EC → ES
D. EC → Controller → EUM → ES
Answer: B — Controller last.

Q9. Secondary 4 days behind — which job, no downtime?

A. ha-failover
B. incremental-replication
C. remove + add-secondary
D. activate-ha-modules
Answer: B — > 7 days needs a revive (primary restarts).

Q10. Which value goes into EUM analytics.accountAccessKey?

A. ad.accountmanager.key.controller
B. appdynamics.es.eum.key
C. Root password
D. EUM license key
Answer: B — Controller → ES uses the other key.

Quick-review cheat sheet

Remember	Value
RBAC model	Users / groups → roles → account + application permissions
Account Owner	Only role for users, groups, roles, authentication provider
Read-Only User	Now called Applications and Dashboards Viewer
Predefined roles	Can't be edited → custom roles
Permission levels	Default → Application → Tier (lower wins)
Multiple roles	A grant in any role wins
LDAP	Group → role mapping · local fallback · keep a local admin
Health URLs	Controller /controller/rest/serverstatus · ES :9080/_ping · EUM /eumaggregator/ping
Health CLI	check-controller-health · show-events-service-health
Admin Console	/controller/admin.jsp · root · Controller Settings / Account Settings
Root password	My Preferences → Change Password · reset via login-db SQL
MySQL root password	EC → Controller Settings → Database Configurations

Sources

- Predefined Roles (on-prem)
- Application Permissions for On-Premises
- LDAP Support
- Update the Root User Passwords
- Start or Stop the Controller (health check CLI)
- Administer the Events Service
- Start and Stop the EUM Server and Database
- Access the Administration Console

Lab — model answers

What a correct result looks like for each Day 8 lab task, with the rule behind it.

1. Custom role "PetClinic Viewer"

1. Settings (gear) → Administration → **Roles** → + → name "PetClinic Viewer".
2. Applications tab: leave default permissions with no View/Edit; for **PetClinic** set Custom → **View** only (Edit and Delete off). Save.
3. **Groups** → + "QA" → assign the PetClinic Viewer role.
4. **Users** → + test user → add to group QA (with LDAP you would map the directory group instead).
5. Log in as the test user: only PetClinic is listed; configuration pages are read-only.

   Only an **Account Owner** (or a role with user administration rights) can do steps 1–4.

2. Combined roles

Yes — the test user can now create and edit PetClinic health rules. When a user holds several roles, a permission granted in any one role takes precedence over its absence in another. They still can't edit other PetClinic configuration that neither role grants.

3. Health rule and email policy

1. Alert & Respond → **Health Rules** → PetClinic → + → type Business Transaction performance, affects all BTs.
2. Critical condition: **Average Response Time > 2000 ms** (static value). Optionally a warning condition, e.g. > 1500 ms. Save.
3. Alert & Respond → **Actions** → + → Send an Email → recipient address.
4. Alert & Respond → **Policies** → + → trigger "Health Rule Violation Started — Critical" for that rule → attach the email action. Save.
5. Generate slow traffic to test; the violation appears and the email is sent (on-prem, the Controller's SMTP settings must be configured).

   Chain to remember: Metric → Health rule violation → Policy → Action.

4. 10-step post-go-live checklist — bank platform

#	Check	Pass criteria
1	Controller via VIP: /controller/rest/serverstatus	200 + available=true on the active node; standby returns 503
2	check-controller-health; log in as admin	No issues; applications visible
3	HA pair	Replication healthy; auto-failover (Watchdog) enabled; planned failover tested in a maintenance window
4	Enterprise Console Platform page	All services running
5	Events Service	show-events-service-health clean; :9080/_ping → 200 on each node and the VIP; Analytics searches return data
6	EUM Server	/eumaggregator/ping → ping; licenses provisioned; browser beacons and mobile data arriving
7	Agents	All expected nodes reporting; agent.log clean; Machine Agents with Server Visibility; DB collectors without errors
8	Licenses	Usage within purchased units
9	Security	CA-signed certificate on s1as, changeit replaced, root/MySQL passwords changed, RBAC roles + LDAP group mapping, local admin kept
10	Operations	Self-monitoring to a separate monitoring Controller; disk latency alert at 3 ms; backups scheduled (levels 1–3); NTP in sync; retention reviewed